ZentroMart VoiceOps agent: capstone report

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LLM: Google Gemini flash-lite (Gemini API) · embeddings: fastembed · top-k 6 · speech: Gnani STT + TTS only · reference date 20 October 2026

Pipeline: customer audio → Gnani STT → agent (LLM + hybrid RAG + order/ticket tools + conversation memory) → Gnani TTS → spoken reply. The same agent object answers typed text, which is how Steps 1 to 3 are evaluated, and voice calls in Step 4.

1. What we built

Step 1: prompt-only baseline. A single LLM call with a system prompt that sets the role (ZentroMart's AI support assistant), the limits (no cancelling, refunding or approving; never share other customers' data; say so when unsure) and the AI disclosure. It has no documents, so it falls back on generic e-commerce knowledge: it guesses plausible but wrong windows and timelines (for example 5 to 7 days for every refund), and it invents policies for questions the documents do not cover.

Step 2: grounded RAG. Each of the 84 Markdown sections becomes one chunk carrying its section ID, doc_type, status and effective_from. Retrieval is hybrid: BM25 for exact terms and SKUs, plus multilingual dense embeddings (paraphrase-multilingual-MiniLM via fastembed) for paraphrases and Hindi questions, fused with reciprocal-rank fusion. An LLM rewrites each message into a standalone English query, and an LLM-generated set of customer-style questions per section (document expansion, cached) bridges the vocabulary gap ("send it back" vs. "reverse pickup"). Explicit cross-references (REF-1, DWM-1) are followed, and a light prior applies the KB-00.1 precedence. Catalog rows are matched by SKU (also spoken SKUs such as "ele zero zero one two") or by product name. The model answers only from what it is given, cites the section IDs it used, uses an exact date_calc tool for business-day maths, and replies "I don't have that information" with an offer of a colleague when nothing covers the question.

Step 3: conversation. A ConversationState keeps the order ID, verification status, items discussed, tickets and the last 12 messages, so follow-ups such as "and the refund?" resolve to the same order and item. The identity check is deterministic code, not left to the model: lookup_order(order_id) runs as soon as an ID is heard, and lookup_order(order_id, value) runs when a 4-digit phone suffix or 6-digit pincode arrives, with one retry and then a lock (IDV-3). The model only sees order data after verification, never sees the verification digits, and verification digits are scrubbed from replies and ticket summaries. Handover calls create_ticket with an ESC-3 summary, and a guard makes sure no reply claims a handover the tool did not confirm (created=True). There is no cancel, refund or credit tool, and the prompt forbids claiming those actions.

Step 4: voice. Gnani STT (REST, format="transcribe" so spoken numbers come back as digits) feeds the unchanged Step 3 agent, and Gnani TTS (timbre-v2.5: Kaveri for English, Nalini for Hindi, Poorvi for Hinglish) speaks the reply. Replies are rewritten for the ear: dates as "the 27th of October 2026", order IDs in groups of four, "Rs 49" as "49 rupees", with no markdown or section IDs. Empty or filler-only transcripts get a short "Sorry, I didn't catch that", and a second miss offers a human colleague. Clips over 55 s are split for the 60 s REST limit. run.py processes a folder of audio, keeps one memory per conversation and saves the transcript, text reply, WAV reply, decision, tool calls, cited sections and all retrieved chunks for each file.

2. Step 1 vs Step 2 on the 52 practice questions

Same scorer for both, following DATA_DICTIONARY.md: a reply is **correct** only if the decision is in acceptable_decisions, every must_include fact is present (alternatives and date formats are equivalent), and no must_not_include phrase appears. Retrieval recall counts gold section IDs and catalog rows found in the model's context; cited recall and precision use the section IDs the agent reported.

Metric	Step 1 prompt-only	Step 2 RAG	Step 2 on ASR text	Step 2 no rewrite
Correct (all checks)	25%	73%	58%	71%
Decision correct	50%	79%	71%	90%
Required facts present	38%	87%	73%	77%
Forbidden claim made	0%	0%	0%	0%
Gold evidence retrieved (recall)	0%	98%	96%	91%
Gold evidence cited (recall)	0%	85%	83%	80%
Cited precision	0%	46%	46%	50%

Step 2 answers **73%** of the questions fully correctly, against **25%** for the prompt-only baseline.

Intent	n	Step 1	Step 2	Intent	n	Step 1	Step 2
cancel_order	2	0%	50%	refund_timeline	14	14%	64%
damaged_item	3	67%	67%	return_eligibility	4	0%	100%
delivery_area	1	100%	100%	return_fee	2	0%	100%
delivery_attempts	1	0%	100%	return_window	11	18%	82%

fraud	1	100%	100%	troubleshooting	1	0%	100%
late_delivery	1	0%	0%	unsupported	5	80%	20%
policy_override	1	100%	100%	warranty	5	0%	100%

Retrieval ablation: each component's contribution, on gold sections other than the always-on agent guides.

Retrieval configuration (no LLM in the loop)	recall@k	hit@k
BM25 only	61%	69%
BM25 + dense	68%	73%
BM25 + dense + doc expansion	87%	91%
full: + LLM query rewrite	98%	100%

3. Step 3 on the 23 practice conversations (96 turns)

Metric	Value
Turns fully correct	73%
Decision correct (answer / clarify / escalate / refuse)	81%
Required facts present	86%
Expected tool calls made, and no unrequested tickets	92%
Forbidden claim made	0%
Gold evidence cited (recall)	62%

Scenario	turns	correct
cancel_shipped_order	8	75%
cancel_unshipped_order	8	88%
compensation_demand	8	50%
express_late_credit	8	100%
failed_verification	8	62%
high_value_damage	4	25%
human_requested	8	88%
refund_status	8	38%
repeat_contact	8	62%
return_eligibility	10	70%
topic_switch	10	100%
where_is_my_order	8	88%

4. What went wrong, and what we changed

• Documents disagree by design: FAQ-1 says refunds take 5 to 7 days, while REF-2 says UPI takes 1 to 3 business days and the Wallet is instant. A model without the documents gives the generic answer, and a weak retriever surfaces the FAQ. Fix: precedence (KB-00.1) is in the system prompt, and each chunk is labelled with its doc_type and status, so the policy wins.

• Date maths: LLMs miscount business days, and this dataset puts Sunday 18 October next to the 19 October warehouse holiday. Fix: an exact date_calc tool, plus a rule that a "by" date is the end of the range.

• Version traps: electronics delivered before 1 August 2026 keep the 10-day window from the superseded v3 policy, and festive-sale fashion gets 30 days. Superseded chunks are kept but labelled, and their scope note is copied onto every chunk.

• Hindi and Hinglish questions against English documents defeat BM25, and so does ASR text with spelled-out numbers. Fix: query rewrite into English, multilingual embeddings, and a spoken-digit normaliser for order IDs, SKUs and phone digits.

• A model can say "I've raised a ticket" without calling the tool. Fix: the decision is checked in code, and if a handover was declared without a successful create_ticket, the code creates the ticket or changes the reply.

Remaining misses in our run (from results/*.jsonl): PQ0004: missing don't have; PQ0009: missing don't have; PQ0015: decision clarify (expected answer); missing don't have; PQ0027: decision clarify (expected answer); PQ0028: decision clarify (expected answer); missing 2026-10-21; PQ0029: decision clarify (expected answer); PC002:2: decision answer (expected clarify); PC002:3: decision escalate (expected answer); tool calls; PC002:4: decision answer (expected escalate); tool calls; PC003:2: decision answer (expected clarify).

Next steps: (1) a cross-encoder reranker to raise precision on the top three chunks; (2) streaming STT/TTS over Gnani's WebSocket API to cut latency on calls; (3) detecting the caller's language from a short first clip and switching STT between en-IN

and hi-IN; (4) an LLM-as-judge pass for paraphrased must_not_include claims that string matching misses.

5. Voice and bonus work

Bonus work: Hindi, Hinglish and English replies with language-matched Gnani voices; a browser demo (app.py) that shows retrieval, tool calls and identity-check state live; a resumable evaluation harness with per-intent and per-language breakdowns and a retrieval ablation; offline tests; and a script that turns the practice set into Gnani TTS audio for end-to-end tests of run.py.

Rules: speech goes through Gnani's API only, with no other STT or TTS anywhere, not even as a fallback. Only the synthetic dataset is used. Keys are read from the environment (.env.example has none). Every test answer is produced by the agent; nothing is hard-coded.